

Cloud Computing (BCS071/BCS713)

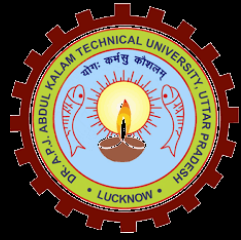
Unit -1 Introduction to Cloud Computing

Unit	Topic
I	Introduction To Cloud Computing: Definition of Cloud – Evolution of Cloud Computing – Underlying Principles of Parallel and Distributed Computing – Cloud Characteristics – Elasticity in Cloud – On- demand Provisioning.

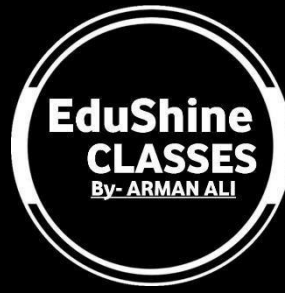
 EduShine Classes – Arman Ali



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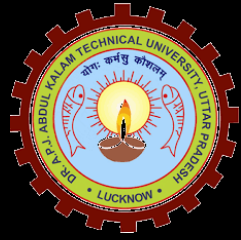
⚡ What is Cloud Computing?

- Cloud computing means **using computer services (like storage, servers, databases, software, and networking) over the Internet** instead of keeping them on your personal computer.
- Cloud computing is like renting computing power, storage, and software over the internet instead of owning and managing your own physical hardware and software

So in simple words:

Cloud = Internet-based computer services.

You just need a good Internet connection, and you can use powerful services from anywhere.



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⚡ Need of Cloud Computing (Why we use it?)

1. Cost Saving 💰

No need to buy expensive hardware/software.
Just pay for what you use (like electricity bill).

2. Scalability 📈

Easily increase or decrease resources.

Example: Amazon adds more servers automatically during Diwali sales.

3. Accessibility 🌐

You can access your files/apps from anywhere in the world, just need the Internet.

Example: Google Drive lets you open files from laptop, phone, or office PC.

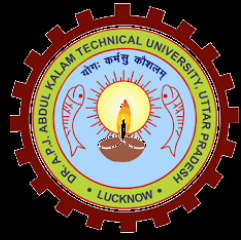
4. Data Backup and Recovery 🔒

Even if your laptop crashes, your files in cloud (like Google Photos or Drive) remain safe.

5. Collaboration 👥

Multiple people can work on the same project at the same time.

Example: In Google Sheets, many friends can edit together.



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6. Automatic Updates ↻

Cloud services get updated automatically (no need to install new versions manually).

✍ Needs of Cloud Computing (Where do we need it?)

1. For Businesses 📁

Startups don't need to buy costly servers, they rent from cloud providers like AWS, Azure, or GCP.

2. For Students/Education 📖

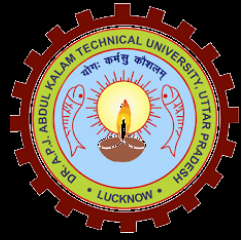
Online classes, assignments, storage of notes, YouTube tutorials all use cloud.

3. For Developers 👤💻

Can test and run apps without buying big servers.

4. For Daily Life 📱

Using Gmail, Google Drive, WhatsApp backup, Netflix – all are examples of cloud computing.



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☁️ Characteristics of Cloud Computing(V.V.IMP)

Cloud computing has some special qualities that make it different from traditional computing. Let's go one by one:

1. On-Demand Self Service

☞ **Means:** You can use cloud services whenever you want, without asking anyone.

💡 **Example:** Just like ordering food from Zomato/Swiggy anytime you want – you don't need to call the restaurant owner; you just click and get it.

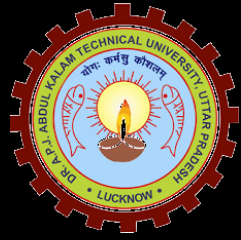
2. Broad Network Access

☞ **Means:** Cloud services can be accessed through the internet from anywhere and from any device (computer, laptop, mobile, tablet).

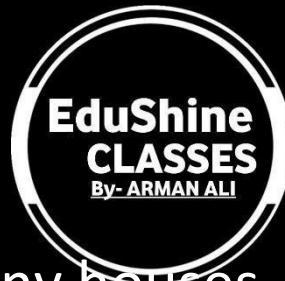
💡 **Example:** You can open **Google Drive** on your phone, laptop, or tablet – doesn't matter where you are.

3. Resource Pooling

☞ **Means:** The cloud provider shares its big resources (like servers, storage, network) with many users. These resources are allocated dynamically depending on who needs them.



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💡 **Example:** Like an electricity board – one power station provides electricity to many houses. You don't know which exact line is giving you power, but you still get it.

4. Rapid Elasticity

👉 **Means:** You can easily increase or decrease your usage of resources. The cloud scales up or down automatically as per your needs.

💡 **Example:** Amazon website gets millions of users during Diwali sale. Cloud automatically provides more servers. After the sale ends, servers are reduced to save cost.

5. Measured Service (Pay-as-you-use)

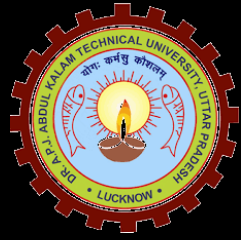
👉 **Means:** You only pay for what you use. The system measures your usage automatically.

💡 **Example:** Like your electricity or mobile data bill – you pay only for the units (GBs or kWh) you use, not for the whole power plant.

7. High Availability and Reliability

👉 **Means:** Cloud is designed to be available 24x7. If one server fails, another one takes its place. So, downtime is very low.

💡 **Example:** Gmail almost never goes down, because Google's cloud system keeps backups ready.



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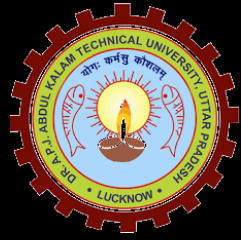
8. Security

🔑 **Means:** Cloud providers give strong security like firewalls, encryption, authentication, so your data stays safe.

💡 **Example:** Like keeping your money in the bank – they have strong vaults and guards. Similarly, the cloud keeps your data safe.

❖ Disadvantages of Cloud Computing :

- Needs internet,
- Risk of security,
- Downtime issues,
- Less control,
- Can be costly,
- Data transfer speed depends on internet,
- Hard to switch providers.



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❖ Cloud Computing Service Provider : (IMP)

Top 10 Cloud Providers





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☁ Evolution of Cloud Computing(V.V.V.IMP)

Cloud Computing did not come suddenly; it evolved slowly with improvements in computing technologies.

1. Distributed Computing (1950s onwards)

Computers were connected together to share tasks.

Work was divided into small parts and executed on different machines.

Goal: Faster problem-solving by sharing resources.

2. Mainframe Computing

Big and powerful computers (mainframes) used by many users through "dumb terminals" (screens without processors).

Goal: Centralized powerful machine serving multiple users.

3. Cluster Computing

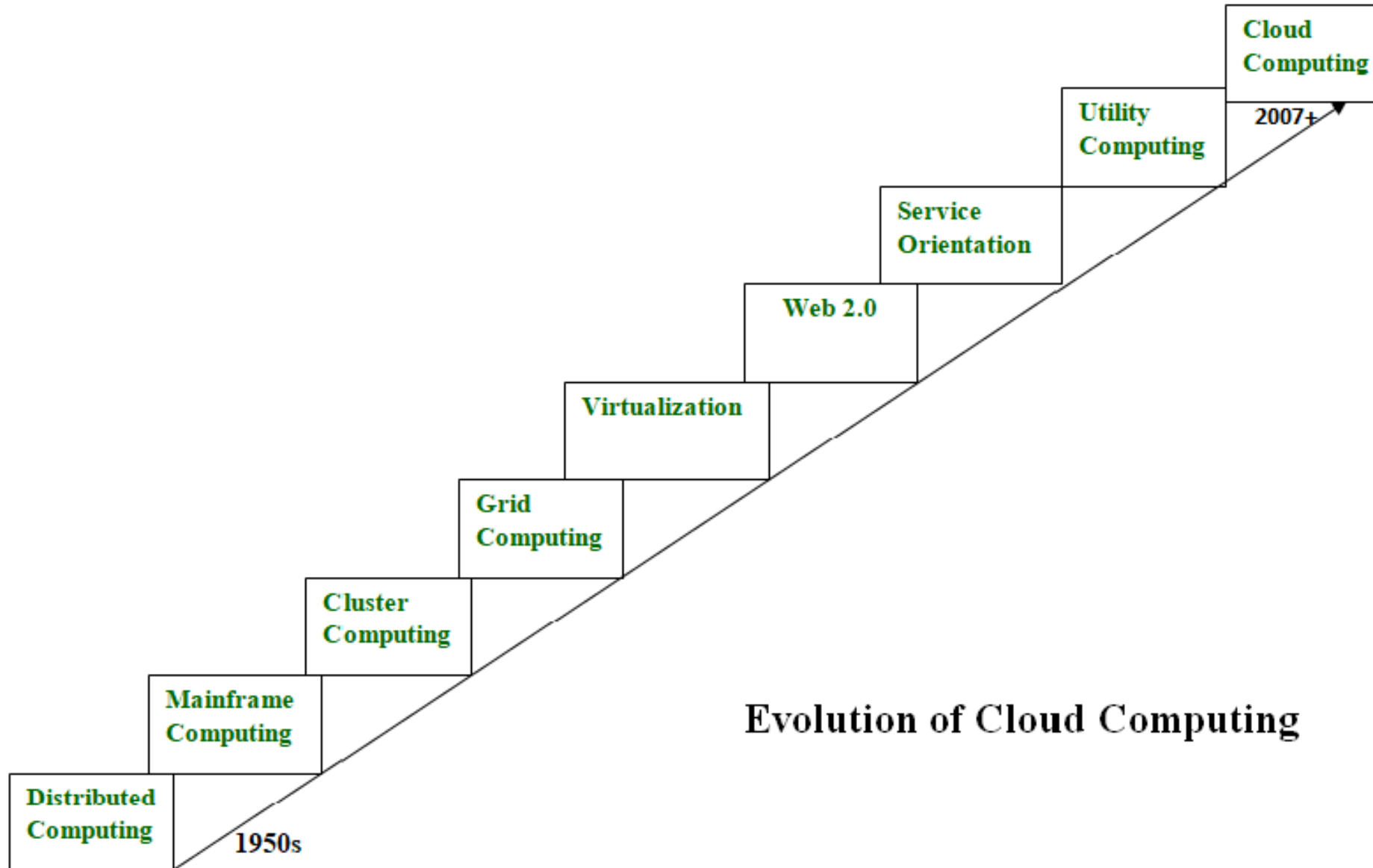
Many computers connected and working as a single powerful system.

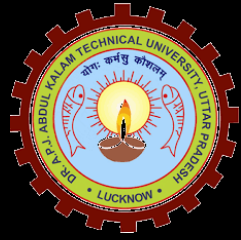
Goal: High performance and reliability at lower cost than a mainframe.





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4. Grid Computing

Resources (computers) from different locations were connected to work together like a "virtual supercomputer".

Used for big scientific research, weather forecasting, etc.

Goal: Use idle resources worldwide efficiently.

5. Virtualization

Multiple virtual machines (VMs) could run on a single physical machine.

Improved hardware utilization, cost saving, and isolation.

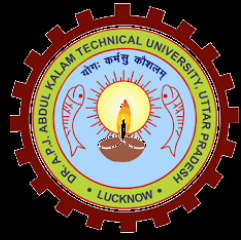
Goal: Better use of one machine for multiple purposes.

7. Service Orientation

Services were delivered as **Software as a Service (SaaS)** (like Gmail, Salesforce).

Instead of buying software, people could use them online.

Goal: Provide services on demand.



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8. Utility Computing

Computing became like electricity → Pay for what you use.

Example: Amazon Web Services (AWS) started offering servers on rent.

Goal: On-demand computing resources at low cost.

9. Cloud Computing (2007 onwards)

All previous ideas merged to form Cloud Computing.

Provides IaaS, PaaS, SaaS on demand.

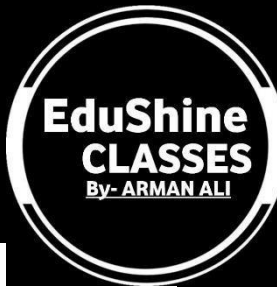
Accessible from anywhere, anytime, with scalability and flexibility.

Goal: Easy access, cost saving, global availability, and elasticity.

Distributed → Mainframe → Cluster → Grid → Virtualization → Web 2.0 → Service Orientation → Utility → Cloud.



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☁ Cloud Computing Architecture(VV.IMP)

Cloud computing has **two main parts**:

1. **Front-end (Client side)**
2. **Back-end (Cloud side)**

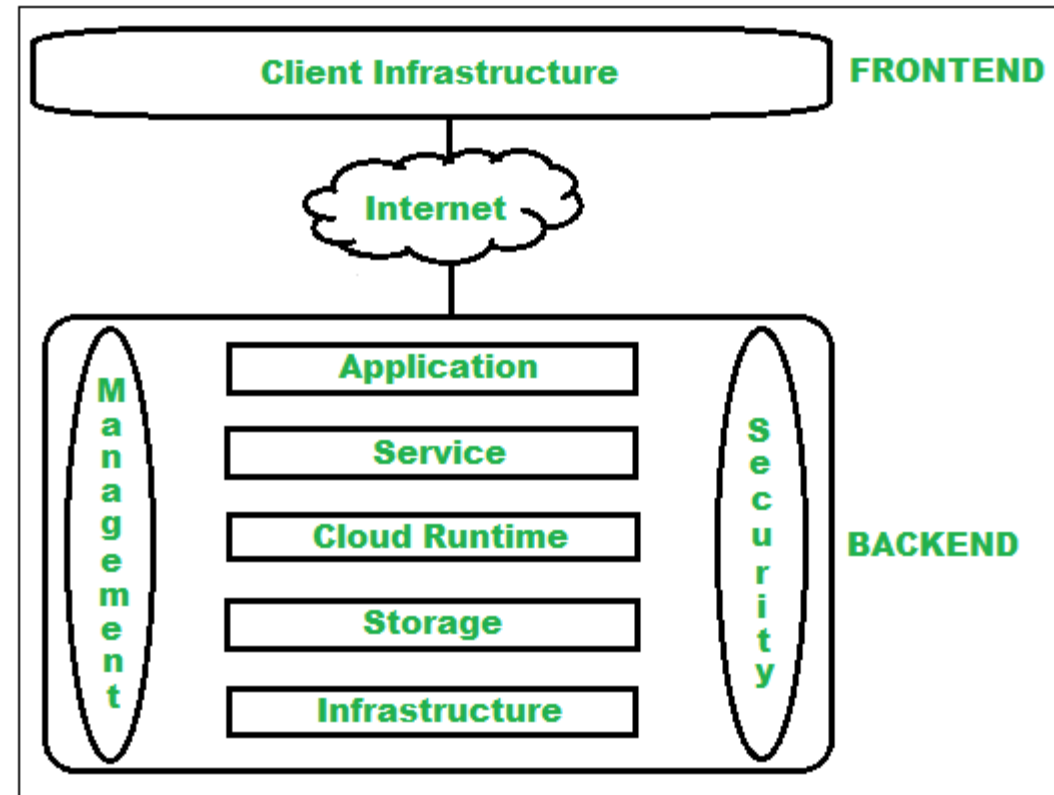
Both are connected through the

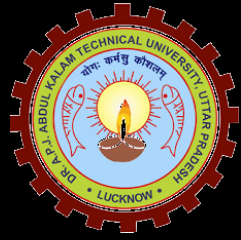
3. **Internet.**

1. Front-end (Client Infrastructure)

This is the **user side**.

Includes the **devices and applications** the user uses to access the cloud (like browsers, laptops, mobiles, thin clients).
Example: You open Gmail using your phone browser → that's the **front-end**.





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2. Internet (Medium)

The internet connects the client (front-end) with the cloud servers (back-end). All communication (requests & responses) happens through the internet.

3. Back-end (Cloud Infrastructure)

This is the **cloud provider side** where actual processing, storage, and service delivery happens. It includes multiple layers:

◆ a) Infrastructure

The base layer → physical hardware like servers, storage devices, and networking.

Example: Data centers that run Google Cloud, AWS, Azure.

◆ b) Storage

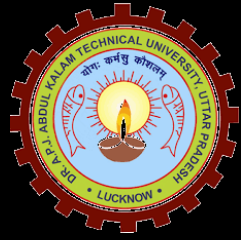
Stores all the cloud data securely.

Scalable → increases storage when demand grows.

Example: Google Drive, Dropbox.

◆ c) Cloud Runtime (Virtualization Layer)

Provides a **virtual environment** where applications run.



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Manages resources like CPU, memory, and storage using virtualization.

Example: VMware, Hyper-V.

◆ d) Services

Provides cloud services like SaaS, PaaS, and IaaS.

Example: Gmail (SaaS), Google App Engine (PaaS), AWS EC2 (IaaS).

◆ e) Application

The actual applications that users access.

Example: Google Docs, Microsoft Office 365, Netflix.

4. Management

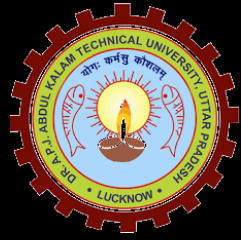
Controls and manages all resources in the cloud (applications, services, runtime, storage, infrastructure).

Ensures smooth functioning and resource allocation.

5. Security

Provides **data protection, authentication, and privacy**.

Very important because cloud is shared among many users.



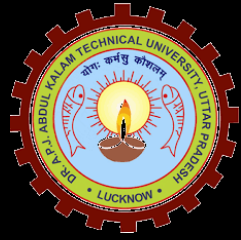
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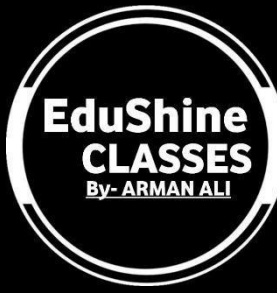
Example: Firewalls, encryption, access control.

❖ Application of Cloud Computing :

- **Data Storage & Backup** – Store files, photos, and videos online (e.g., Google Drive, Dropbox).
- **Web Hosting** – Hosting websites and applications on cloud servers.
- **Software as a Service (SaaS)** – Using apps like Gmail, MS Office 365 without installing.
- **E-commerce** – Online shopping platforms like Amazon, Flipkart use cloud.
- **Social Networking** – Facebook, Instagram, Twitter run on cloud servers.
- **Education** – Online classes, e-learning platforms (Coursera, Google Classroom).
- **Banking & Finance** – Online transactions, mobile banking apps.
- **Entertainment** – OTT platforms like Netflix, Hotstar use cloud for video streaming.
- **Business Applications** – CRM tools, project management, and collaboration (Zoom, Slack).
- **Healthcare** – Storing patient records and telemedicine services.



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❖ What is Parallel Computing?(V.VVIMP)

Parallel computing means **dividing a big problem into smaller parts**.

These smaller parts are solved **at the same time** (simultaneously) using **multiple processors (CPUs/cores)**.

Finally, the results are combined to get the **final output**.

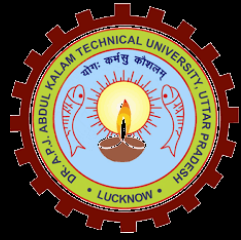
☞ **Example:** Imagine you want to wash 100 clothes.

Normal way (sequential): 1 person washes all clothes one by one.

Parallel way: 4 people wash 25 clothes each at the same time → work finishes faster.

➤ Steps in Parallel Computing :

1. **Problem** – A large task that needs to be solved.
2. **Divide into Instructions** – Break the problem into smaller instructions or tasks.
3. **Assign to Processors** – Each processor gets a part of the work.
4. **Execute Simultaneously** – All processors work at the same time.
5. **Combine Results** – The outputs from all processors are merged into the final solution.



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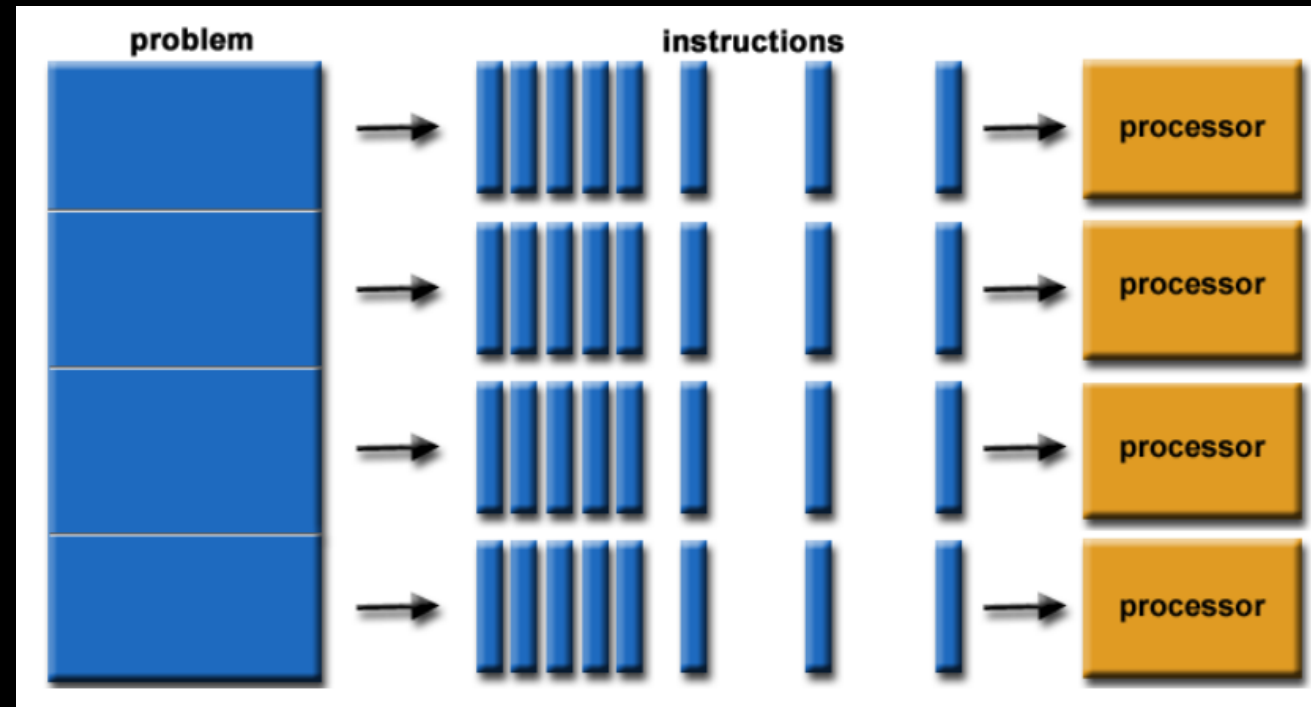


◆ Advantages of Parallel Computing over Serial Computing

- ✓ Saves **time and money** (work is done faster).
- ✓ Can solve **big and complex problems** (serial computing fails here).
- ✓ Uses **resources more effectively** (no idle CPU parts).
- ✓ Allows use of **distributed resources** (not only local machines).

❖ Disadvantages :

- Difficult to divide some problems.
- Requires special hardware/software.
- Costly setup if many processors are needed.





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❖ ♦ Types of Parallelism

1. Bit-level Parallelism

Processor handles more bits at a time.

Example: An 8-bit processor adds a 16-bit number in 2 steps, but a 16-bit processor does it in 1 step.

2. Instruction-level Parallelism

Processor rearranges and executes multiple instructions in one clock cycle (without changing result).

3. Task Parallelism

A big task is divided into smaller subtasks.

Subtasks run at the same time on different processors.

4. Data Parallelism (DLP)

Same instruction is applied on multiple data sets at once.

Example: Adding two arrays by splitting data among processors.



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2. Distributed Computing

Distributed computing is a system where **many computers (or machines) work together to solve a single big problem.**

Instead of one computer doing all the work, the work is **divided (distributed)** among many computers connected through a network.

✓ Main Idea:

- A big task is **broken into small parts.**
- These parts are given to different computers.
- All computers work at the same time (parallel).
- At the end, results are combined to get the final answer.

❖ Key Features:

1. **Multiple Computers** – connected by a network.
2. **Shared Workload** – tasks are divided among them.
3. **Faster Processing** – work is done quickly.
4. **Fault Tolerance** – if one machine fails, others can continue.



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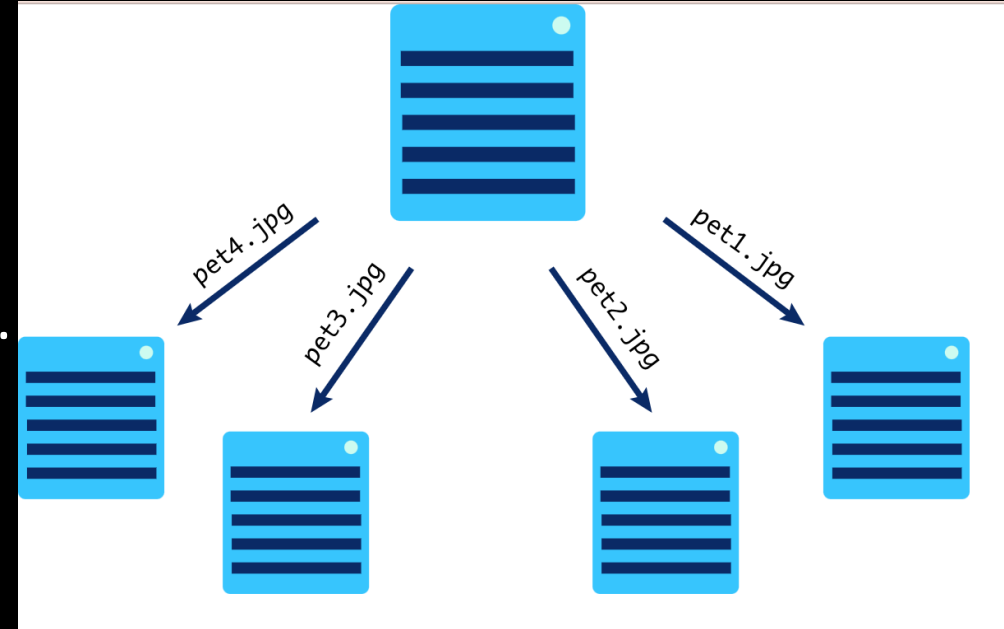
Advantages:

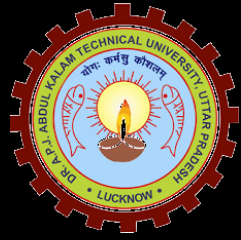
- Saves **time** (parallel processing).
- Uses resources more efficiently.
- Can handle **big problems** that one computer cannot.

Simple Real-Life Examples:

1. **Google Search** – thousands of servers work together.
2. **Banking Systems** – transactions handled by multiple computers.
3. **Weather Forecasting** – uses many supercomputers together.

Distributed computing is a model where many computers work together on a problem by dividing tasks among them and connecting through a network. It improves speed, resource sharing, fault tolerance, and can solve very large problems. Example: Google search engine and weather forecasting use distributed computing.



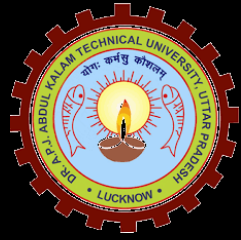


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❖ Difference between Parallel Computing Vs Distributed Computing :

S.No.	Parallel Computing	Distributed Computing
1	Many operations are performed simultaneously.	System components are located at different locations.
2	Single computer is required.	Uses multiple computers.
3	Multiple processors perform multiple operations.	Multiple computers perform multiple operations.
4	It may have shared or distributed memory.	It only has distributed memory.
5	Processors communicate via a bus.	Computers communicate through message passing.
6	Improves system performance.	Enhances system scalability, fault tolerance, and resource sharing.



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☁ Cloud Elasticity

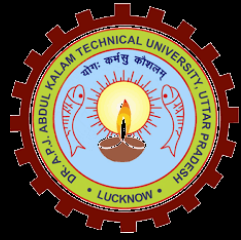
- Elasticity means **automatic adjustment of resources** (CPU, RAM, storage, bandwidth) depending on sudden demand.
- When more people use your app/website → cloud **adds extra resources automatically**.
- When fewer people use → cloud **removes extra resources automatically**.
- Saves **money** because you only pay for what you use (pay-per-use).
- Best for **temporary / seasonal demand**.

👉 Example:

An online shopping site gets huge traffic during **Christmas sales**. Elasticity helps it automatically add more servers during sales and remove them when sales end

☁ Cloud Scalability

Scalability means **growing resources steadily (manually or automatically)** to handle **permanent increase in workload**.



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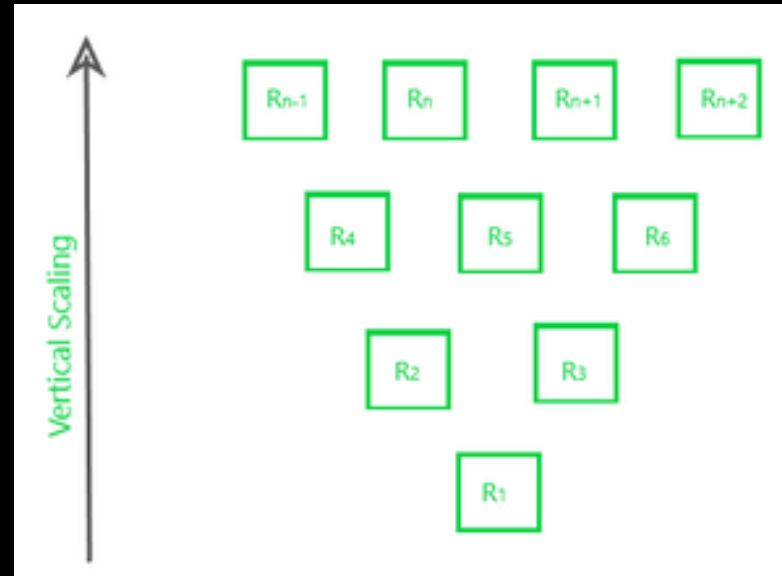
- It is **planned growth**, not sudden like elasticity.
- Best for **long-term resource needs**.

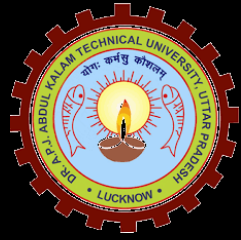
➡ Example:

A company's **database** keeps growing every year as business expands. They **scale up** storage and computing power to handle the larger workload.

↻ Types of Scalability

1. **Vertical Scalability (Scale Up):** Increase the power of existing machine (e.g., add more RAM or CPU to the same server).

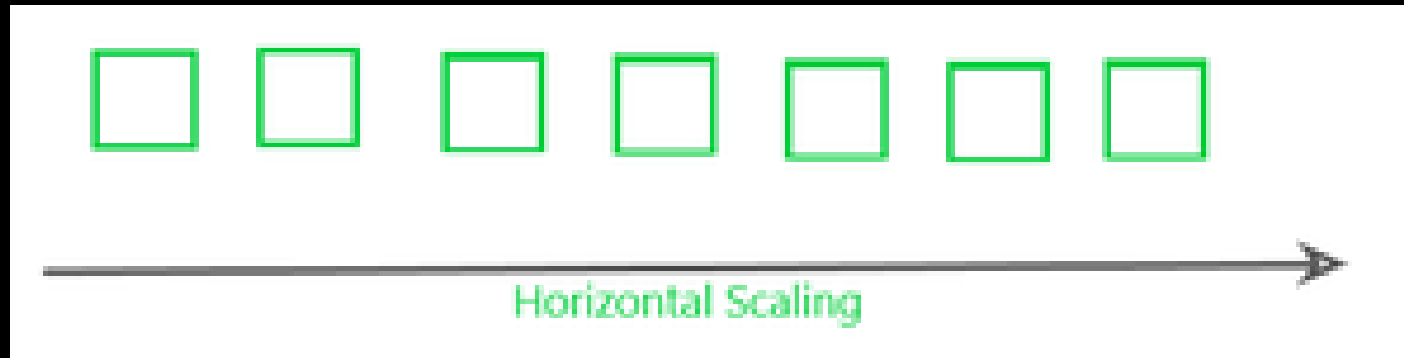




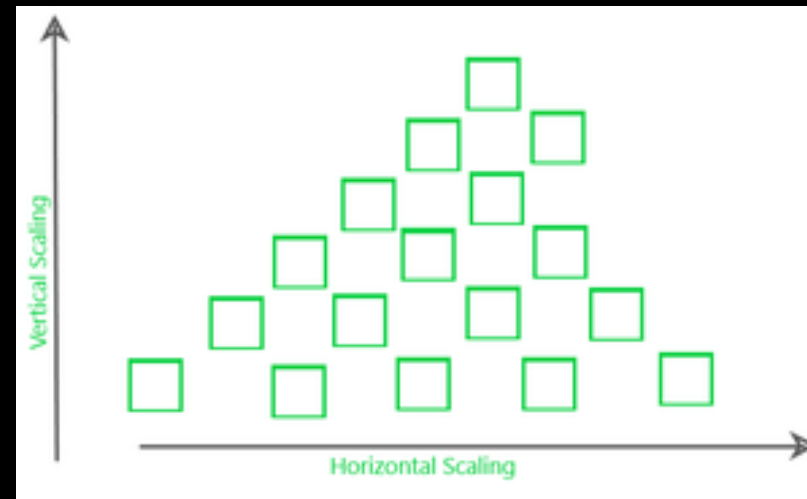
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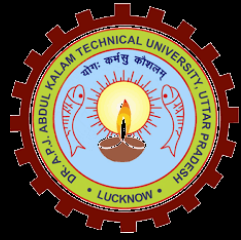


2. Horizontal Scalability (Scale Out): Add more machines/servers to share the load.

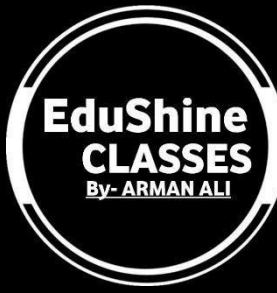


3. Diagonal Scalability: A mix of both vertical + horizontal scaling.





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❖ Difference between Elasticity & Scalability : (IMP)

Elasticity

Handles **sudden increase or decrease** in workload.

Dynamic (resources go up & down automatically).

Used by **small/medium businesses** with seasonal demands.

Short-term solution.

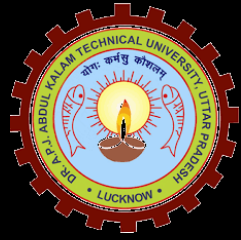
Scalability

Handles **steady, predictable increase** in workload.

Static/Planned (resources are increased when needed permanently).

Used by **big companies** with continuously growing users.

Long-term solution.



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❖ On-Demand Provisioning in Cloud Computing :

On-demand provisioning means getting computing resources (like servers, storage, or applications) whenever you need them, and releasing them when you don't.

It is just like ordering electricity, water, or food online – you get it instantly when required, and stop when you don't need it.

1 .Traditional Way (Before Cloud):

- Companies had to buy and install their own servers.
- Even if they used them for 2 hours only, they had to pay for the whole machine.
- This was expensive and time-wasting.

2. With On-Demand Provisioning (Cloud):

- You don't buy hardware.
 - You just request resources (CPU, storage, database, etc.) from the cloud.
 - Cloud provider (like AWS, Azure, Google Cloud) gives you resources within seconds.
 - You pay only for what you use.
-



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❖ Key Features :

- **Instant availability** – Resources are ready quickly.
- **Scalability** – Increase or decrease resources anytime.
- **Pay-as-you-go** – No need to spend extra money.
- **Flexibility** – Use resources only when needed.

Example

- Imagine you are building a website.
- During normal days → You need only 2 servers.
- During festival season (Diwali sales) → You suddenly need 20 servers.
- With on-demand provisioning → You can instantly add 18 extra servers and remove them later when traffic reduces.

On-demand provisioning in cloud computing means **providing IT resources (like servers, storage, databases) instantly whenever needed, instead of pre-installing them.** It saves **time, money, and effort**, and allows **scaling up and down** as per demand.



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Thank You....



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